Type MS · Area of a triangle using half ab sin C

Answer 1 of 12 · Question 1 part whole · 4 marks

2024 JUN · Paper 01 · extract 1 of 5 · find part whole in this mark scheme extract

1	**Angle** $(4x)^2 = (2x)^2 + (3x)^2 - 2\times 2x\times 3x\times \cos ABC$ **or** $(\cos ABC =)\dfrac{(2x)^2+(3x)^2-(4x)^2}{2\times 2x\times 3x} \Rightarrow \angle ABC = 104.47751...^{o}\,(1.823476...^{c})$ OR $(3x)^2 = (2x)^2 + (4x)^2 - 2\times 2x\times 4x\times \cos BAC$ **or** $(\cos BAC =)\dfrac{(2x)^2+(4x)^2-(3x)^2}{2\times 2x\times 4x} \Rightarrow \angle BAC = 46.56746...^{o}\,(0.812755...^{c})$ OR $(2x)^2 = (3x)^2 + (4x)^2 - 2\times 3x\times 4x\times \cos ACB$ **or** $(\cos ACB =)\dfrac{(3x)^2+(4x)^2-(2x)^2}{2\times 3x\times 4x} \Rightarrow \angle ACB = 28.95502...^{o}\,(0.505360....^{c})$ **Area** $50 = \dfrac{1}{2}\times 2x\times 3x\times \sin"104.47751"^{o} \Rightarrow x^2\,(=17.213259...)$ $\Rightarrow x = 4.15$ OR $50 = \dfrac{1}{2}\times 2x\times 4x\times \sin"46.56746"^{o} \Rightarrow x^2\,(=17.213259...)$ $\Rightarrow x = 4.15$ OR $50 = \dfrac{1}{2}\times 3x\times 4x\times \sin"28.95502"^{o} \Rightarrow x^2\,(=17.213259...)$ $\Rightarrow x = 4.15$	M1A1 [M1A1] [M1A1] dM1 A1 [dM1 A1] [dM1 A1] [4]
ALT1	Uses Heron's formula $s = \dfrac{2x+3x+4x}{2} = \dfrac{9x}{2}$ oe $50 = \sqrt{"\frac{9x}{2}"\left("\frac{9x}{2}"-2x\right)\left("\frac{9x}{2}"-3x\right)\left("\frac{9x}{2}-4x"\right)} = \sqrt{\dfrac{135x^4}{16}}$ $\Rightarrow x^2 = 17.213259... \Rightarrow x = 4.15$	M1A1 dM1 A1 [4]

Type MS · Area of a triangle using half ab sin C

Answer 1 of 12 · Question 1 part whole · 4 marks

2024 JUN · Paper 01 · extract 2 of 5 · find part whole in this mark scheme extract

ALT 2	$(4x)^2=(2x)^2+(3x)^2-2\times 2x\times 3x\times\cos ABC$ **or**	
	$(\cos ABC=)\dfrac{(2x)^2+(3x)^2-(4x)^2}{2\times 2x\times 3x}\left(=-\dfrac{1}{4}\right)$	M1
	$\Rightarrow(\sin ABC=)\sqrt{\dfrac{15}{16}}\left(=\dfrac{\sqrt{15}}{4}=0.9682458...\right)$ oe	A1
	OR	
	$(3x)^2=(2x)^2+(4x)^2-2\times 2x\times 4x\times\cos BAC$ **or**	{M1
	$(\cos BAC=)\dfrac{(2x)^2+(4x)^2-(3x)^2}{2\times 2x\times 4x}\left(=\dfrac{11}{16}\right)$	
	$\Rightarrow(\sin ABC=)\sqrt{\dfrac{135}{256}}\left(=3\dfrac{\sqrt{15}}{16}=0.7261843...\right)$ oe	A1}
	OR	
	$(2x)^2=(3x)^2+(4x)^2-2\times 3x\times 4x\times\cos ACB$ **or**	{M1
	$(\cos ACB=)\dfrac{(3x)^2+(4x)^2-(2x)^2}{2\times 3x\times 4x}\left(=\dfrac{7}{8}\right)$	
	$\Rightarrow(\sin ABC=)\sqrt{\dfrac{15}{64}}\left(=\dfrac{\sqrt{15}}{8}=0.4841229...\right)$ oe	A1}
	$50=\dfrac{1}{2}\times 2x\times 3x\times"\sqrt{\dfrac{15}{16}}"\Rightarrow x^2\left(=17.213259...\right)$ or $x=$	dM1
	$\Rightarrow x=4.15$	A1
	OR	
	$50=\dfrac{1}{2}\times 2x\times 4x\times"\sqrt{\dfrac{135}{256}}"\Rightarrow x^2\left(=17.213259...\right)$ or $x=$	{dM1
	$\Rightarrow x=4.15$	A1}
	OR	
	$50=\dfrac{1}{2}\times 3x\times 4x\times"\sqrt{\dfrac{15}{64}}"\Rightarrow x^2\left(=17.213259...\right)$ or $x=$	{dM1
	$\Rightarrow x=4.15$	A1}
		[4]
		Total 4 marks

Type MS · Area of a triangle using half ab sin C

Answer 1 of 12 · Question 1 part whole · 4 marks

2024 JUN · Paper 01 · extract 3 of 5 · find part whole in this mark scheme extract

Mark	Notes
M1	For a fully correct substitution into the cosine **formula** as shown. ie including an equals sign. OR for a fully correct **expression** for cos*ABC* or cos*BAC* or cos*ACB* Allow students to use just *B*, *A* and *C* for angles or any labelling.
A1	For one of the correct angles in the triangle.
Note for M1 A1 In this question, we will override the general principle of marking for multiple attempts, if the attempts are finding other angles. Mark the attempt that is correct. The *x* can be consistently omitted **or** the *x* can be recovered (also indicated by a correct angle) **or** values in the correct proportions can be used as an alternative (also indicated by a correct angle). M1 may also be awarded if the values in the expression or equation lead to the correct angle for those values, regardless of the labelling of the angle. This may also lead to A1	

Type MS · Area of a triangle using half ab sin C

Answer 1 of 12 · Question 1 part whole · 4 marks

2024 JUN · Paper 01 · extract 4 of 5 · find part whole in this mark scheme extract

ALT2	
M1	For a fully correct substitution into the cosine **formula** as shown. ie including an equals sign. OR for a fully correct **expression** for cos*ABC* or cos*BAC* or cos*ACB* Allow students to use just *B*, *A* and *C* for angles or any labelling.
A1	For one of the correct values for sin(relevant angle)
Note for M1 A1 The x can be consistently omitted **or** the x can be recovered (also indicated by a correct value for sin) **or** values in the correct proportions can be used as an alternative (also indicated by a correct value for sin). M1 may also be awarded if the values in the expression or equation lead to the correct sin value for those values, regardless of the labelling of the angle. This may also lead to A1 $(2x)^2 = (3x)^2 + (4x)^2 - 2\times 3x\times 4x\times \cos A$ **or** e.g. $(\cos A =)\dfrac{(3x)^2+(4x)^2-(2x)^2}{2\times 3x\times 4x} \Rightarrow \sin A = \sqrt{\dfrac{15}{64}}$ All the values here are correct to give $\Rightarrow (\sin ABC =)\sqrt{\dfrac{15}{64}}$, but the labelling of angle A is incorrect. Allow values to be rounded to 1 dp.	
dM1	For using the correct formula for the area of a triangle using 50 and their sin value correctly and the correct 2 sides for their angle and a fully correct rearrangement to find x^2 or x If M1 has been gained but their sin value is incorrect, students must be seen to have substituted into $\cos^2(\text{angle}) + \sin^2(\text{angle}) = 1$. Though a poor rearrangement to give an incorrect sin value, will potentially allow this mark to be awarded. Sight of 17.2(1…….) following M1 A1 will imply this mark. Dependent on previous method mark.
A1	For the correct value of x. Accept awrt 4.15 [cm]

Type MS · Area of a triangle using half ab sin C

Answer 1 of 12 · Question 1 part whole · 4 marks

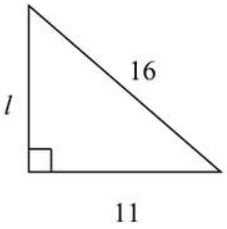
2024 JUN · Paper 01 · extract 5 of 5 · find part whole in this mark scheme extract

Question	Scheme	Marks
For part (a) of this question, mark using the scheme which gives the most marks.		

Type MS · Area of a triangle using half ab sin C

Answer 2 of 12 · Question 2 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
2 (a)	$\cos ABC = \dfrac{(2x)^2+(4x)^2-(3x)^2}{2\times 2x\times 4x} = \dfrac{x^2(4+16-9)}{x^2(16)} = \dfrac{11}{16}$	M1A1
	 $l = \sqrt{16^2-11^2} = 3\sqrt{15}$	M1
	$\sin ABC = \dfrac{3\sqrt{15}}{16}$ *	A1 [4]
	ALT	
	$\sin^2 ABC = 1 - \dfrac{121}{256} = \dfrac{135}{256}$ Þ $\sin ABC = \dfrac{3\sqrt{15}}{16}$ *	{M1A1}
(b)	$\dfrac{75\sqrt{15}}{64} = \dfrac{1}{2}\times 2x\times 4x\times \dfrac{3\sqrt{15}}{16} \Rightarrow x^2 = \dfrac{25}{16} \Rightarrow x = \dfrac{5}{4}$ oe (positive root only)	M1A1 [2]
		Total 6 marks

Type MS · Area of a triangle using half ab sin C

Answer 2 of 12 · Question 2 part b · 2 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

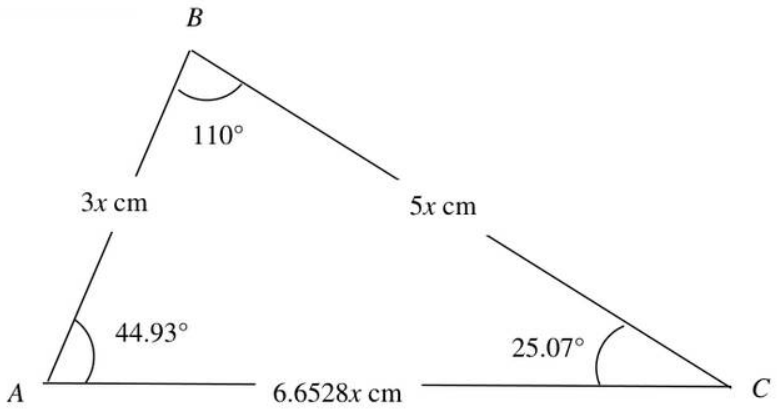
Type MS · Area of a triangle using half ab sin C

Answer 3 of 12 · Question 2 part b · 3 marks

2023 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

2(a)	$AC = \sqrt{(3x)^2 + (5x)^2 - 2\times 3x\times 5x\times\cos 110^\circ} = 6.6528...x$ $\sin\angle BCA = \frac{3x\sin 110^\circ}{'6.6528'x} = 0.4237... \Rightarrow \angle BCA = 25.07...^\circ$ Accept awrt 25.1° **ALT** (for 2nd M1 A1) $\cos\angle BCA = \frac{(5x)^2 + ('6.6528'x)^2 - (3x)^2}{2\times 5x\times '6.6528'x} = 0.9057... \Rightarrow \angle BCA = 25.07^\circ$ Accept awrt 25.1°	M1A1 M1 A1 [4] [M1 A1]
(b)	$24 = \frac{1}{2}\times 3x\times 5x\times\sin 110^\circ$ $\Rightarrow x = \sqrt{\frac{24\times 2}{3\times 5\times\sin 110^\circ}} = (1.84536...)$ Accept awrt 1.85	M1 M1 A1 [3]
		Total 7 marks

Useful sketch



Type MS · Area of a triangle using half ab sin C

Answer 3 of 12 · Question 2 part b · 3 marks

2023 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using a correct cosine rule for AC or AC^2. Note, this is given in the formula sheet so must be correct for this mark.
	A1	For finding a length of $AC = 6.6528...x$ or $AC = \sqrt{44.26....x^2}$ accept awrt $6.7x$ or $\sqrt{44.3x^2}$ **Note:** Allow the missing x here and throughout their working This mark can be implied by sight of $AC = \sqrt{34x^2 - 30x^2\cos 110}$ if they carry it through to find the angle in the next step. For example: $\sin\angle BCA = \dfrac{3x\sin 110^\circ}{\sqrt{34x^2 - 30x^2\cos 110}}$
	M1	For using any appropriate trigonometry to find the size of angle BCA. For example; **Sine Rule** $\sin\angle BCA = \dfrac{3x\sin 110^\circ}{'6.6528'x} = 0.4237... \Rightarrow \angle BCA = 25.07...^\circ$ **Cosine Rule** $\cos\angle BCA = \dfrac{(5x)^2 + (6.6529x)^2 - (3x)^2}{2\times 5x\times 6.6529x} \Rightarrow \angle BCA = 25.07$ Allow a missing x from their working provided it is consistent. Do not allow for example $\sin\angle BCA = \dfrac{3\boldsymbol{x}\sin 110^\circ}{'6.6528'}$
	A1	For awrt 25.1°
	ALT – Uses sine rule	
	M1	Use of sine rule: $\dfrac{\sin(70-\theta)}{5x} = \dfrac{\sin\theta}{3x} \Rightarrow \left[\dfrac{\sin(70-\theta)}{\sin\theta} = \dfrac{5}{3}\right]$
	A1	Expands $\sin(70-\theta) = \sin 70\cos\theta - \cos 70\sin\theta$
	M1	Uses the tan identity and rearranges to make $\tan\theta$ the subject. $\dfrac{\sin 70\cos\theta - \cos 70\sin\theta}{\sin\theta} = \dfrac{5}{3}$ $\Rightarrow \dfrac{\sin 70}{\tan\theta} - \cos 70 = \dfrac{5}{3} \Rightarrow \tan\theta = \dfrac{\sin 70}{\cos 70 + \dfrac{5}{3}}$
	A1	For awrt 25.1°
(b)	M1	For using the correct formula for the area of a triangle with the correct angle and side lengths. e.g., Can use AB and AC with $\angle ABC = 110^\circ$ or BC and AC with their $\angle BCA$ [25.1°] or AB and AC with their $(180^\circ - 110^\circ - 25.07^\circ = 44.93^\circ)$

Type MS · Area of a triangle using half ab sin C

Answer 3 of 12 · Question 2 part b · 3 marks

2023 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

	M1	For finding a value of x using their value for AC and BCA or BAC The processing must be correct for this mark and they must be finding the square root of x for this mark.
	A1	For awrt 1.85

Question	Scheme	Marks

Type MS · Area of a triangle using half ab sin C

Answer 4 of 12 · Question 3 part b · 4 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

3 (a)	$\cos(\angle B) = \frac{10^2+8^2-12^2}{2\times10\times8} = \frac{10^2+6^2-x^2}{2\times10\times6}$	M1 M1
	$\frac{20}{160} = \frac{136-x^2}{120}$	
	$x^2 = 121$	dM1
	$x = 11$ *	A1 cso (4)
	Alternative Method	
	$\cos(ADB) = \frac{x^2+6^2-10^2}{2\times6\times x} = \frac{x^2-64}{12x}$	M1
	$\cos(ADC) = \frac{x^2+2^2-12^2}{2\times2\times x} = \frac{x^2-140}{4x}$	M1
	$\frac{x^2-64}{12x} + \frac{x^2-140}{4x} = 0$	
	$4x^2 = 484$	dM1
	$x = 11$ *	A1 cso (4)
(b)	Angle ABC / $ABD = \cos^{-1}\left(\frac{10^2+8^2-12^2}{2\times10\times8}\right) = 82.8^\circ$	B1 M1
	$\frac{1}{2}\times10\times6\times\sin82.8 = 29.8$	M1 A1 (4)
		[8]
	Alternative Method 1	
	Angle $ADB = \cos^{-1}\left(\frac{11^2-64}{12\times11}\right) = 64.42^\circ$	B1 M1
	$\frac{1}{2}\times6\times11\times\sin64.42 = 29.8$	M1 A1 (4)
	Alternative Method 2	
	$\cos ABC = \frac{10^2+8^2-12^2}{2\times10\times8} = \frac{1}{8}$	B1
	$\sin ABC$ / $ABD = \sqrt{1-\left(\frac{1}{8}\right)^2} = \sqrt{\frac{63}{64}} = \frac{3}{8}\sqrt{7}$	M1
	$\frac{1}{2}\times10\times6\times\frac{3}{8}\sqrt{7} = 29.8$	M1 A1 (4)

Type MS · Area of a triangle using half ab sin C

Answer 4 of 12 · Question 3 part b · 4 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

	Notes
(a)	
M1	Use of cosine rule to obtain a correct expression for $\cos(\angle B)$. The correct formula in either form may be used.
M1	Use of cosine rule to obtain a second correct expression for $\cos(\angle B)$. The correct formula in either form may be used.
dM1	Dependant on previous M mark - for solving leading to $x^2 = ...$
A1 cso	For obtaining the given result
	Alternative
M1	Use of cosine rule to obtain a correct expression for $\cos(ADB)$. The correct formula in either form may be used.
M1	Use of cosine rule to obtain a correct expression for $\cos(ADC)$. The correct formula in either form may be used.
dM1	Dependant on previous M mark - for solving leading to $4x^2 = ...$
A1	For obtaining the given result
(b)	
B1	For use of the cosine rule to find angle B
M1	For $\cos^{-1}\left(\frac{10^2+8^2-12^2}{2\times10\times8}\right)$ oe
M1	Use of $\frac{1}{2}ab\sin C$ (correct for their angle)
A1	29.8
	Alternative 1
B1	For use of the cosine rule to find angle ADB
M1	For $\cos^{-1}\left(\frac{11^2-64}{12\times11}\right)$ oe
M1	Use of $\frac{1}{2}ab\sin C$ (correct for their angle)
A1	29.8
	Alternative 2
B1	For use of the cosine rule to find cos B
M1	Use of $\sin^2 A + \cos^2 A = 1$
M1	Use of $\frac{1}{2}ab\sin C$ (correct for their angle)
A1	29.8

Type MS · Area of a triangle using half ab sin C

Answer 4 of 12 · Question 3 part b · 4 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle using half ab sin C

Answer 5 of 12 · Question 3 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 5 · find part a in this mark scheme extract

		$\left(-\frac{}{3n}\right) \Rightarrow \left(\frac{}{9n}\right)$ or $\left(-\frac{}{3n}\right) \Rightarrow -\left(\frac{}{9n^2}\right)$ then this is M0 Their equation must be either a quadratic or linear if they cancel an n This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$ isw $n = 0$ seen. This is equivalent to cancelling n's
	A1	For the correct value of n **and** A
	ALT 1	**First two marks same as main method**
	M1	Substitutes n in terms of A into an equation in A $\frac{'-\frac{1}{3A}'\left('-\frac{1}{3A}'-1\right)(A)^2}{2}=\frac{5}{36}$ **Do not** allow a substitution of $n=-\frac{1}{3}$ for this mark
	dM1	For solving **their** equation in A to find a value for A Their equation must be either a quadratic or linear if they cancel an A This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$
	A1	For the correct value of n **and** A
	ALT 2	**First two marks same as main method**
	M1	Substitutes nA into $\frac{(nA)^2-A(An)}{2!}=\frac{5}{36}$
	dM1	Solves their equation to find a value for A provided no errors introduced. This is dependent on the previous M mark only.
	A1	For the correct value of n **or** A $\quad n=-\frac{2}{3}, \quad A=\frac{1}{2}$
	A1	For the correct value of n **and** A

Type MS · Area of a triangle using half ab sin C

Answer 5 of 12 · Question 3 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 5 · find part a in this mark scheme extract

(b)	M1	Uses the **correct form** for the fourth term of a binomial expansion with their A and their n. You may see this in terms of A and n in part (a) Accept as a minimum, the correct power of $\left(\frac{1}{2}\right)^3$ with the correct denominator i.e. 3! Allow the presence of x^3 for this mark.
	A1	For the correct value of $-\frac{5}{81}$ If you see the correct value following the correct values of A and n with no working, award M1A1

Type MS · Area of a triangle using half ab sin C

Answer 5 of 12 · Question 3 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks
3(a)	$36=\frac{1}{2}\times x\times(x+7)\times\sin 30^\circ \Rightarrow 36=\frac{x(x+7)}{4}\Rightarrow x^2+7x-144=0$	M1
	[Accept also e.g., $x^2+7x=144$]	
	$(x-9)(x+16)=0\Rightarrow x=9,(-16)\,(\text{cm})$*	M1A1 cso [3]
(b)	$BC=\sqrt{9^2+16^2-2\times9\times16\cos30^\circ}=9.3586...\approx9.36\,(\text{cm})$	M1A1 [2]
(c)(i)	$\frac{\sin ABC}{16}=\frac{\sin30^\circ}{9.3586}\Rightarrow\angle ABC=180^\circ-58.7408...\approx121.3^\circ$	M1A1
(ii)	$\angle BCA=180^\circ-30^\circ-121.259^\circ=28.740...\approx28.7^\circ$	B1ft [3]
		Total 8 marks

Type MS · Area of a triangle using half ab sin C

Answer 5 of 12 · Question 3 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 4 of 5 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	States or uses $\theta = \frac{21-r}{r}$
	M1	Uses the correct formula for the area of a sector and substitutes their expression for θ
	A1 cso	For the correct expression for A exactly as written with no errors. That is $A = \frac{r}{2}(21-r)$ including $A =$.. However, if they have a string of working, then allow $A =$ at the top For example ., A = ... =.... = ... = $\frac{r}{2}(21-r)$ is acceptable
	ALT	
	B1	States $l = 21-r$
	M1	Uses the alternative formula $A = \frac{1}{2}rl$ and substitutes in their l
	A1 cso	For the correct expression for A exactly as written with no errors.
(b)	B1	Forms a correct inequality **and** a correct 3TQ (accept in any form) using the given expression for A and the value of 27
	M1	Solves their 3TQ by any method to obtain two values for r If they use a calculator to solve an incorrect 3TQ, award only if a method is seen. If the 3TQ is correct, and they use a calculator, award for correct roots seen.
	M1	Forms an inequality for the **inside** region using their values of r Allow any critical values which were obtained from their quadratic equation. Accept any acceptable form that defines a continuous **inside** region For example: $3 \leqslant r$ and $r \leqslant 18$ or $3 \leqslant r \cap r \leqslant 18$ Accept even $3 \leqslant r$ or $r \leqslant 18$
	A1	For the correct inequality only in this form. $3 \leqslant r \leqslant 18$
(c)	B1ft	For both $\theta = \frac{1}{6}$ and $\theta = 6$ You must follow through their values for r for this mark.
	B1	For the correct inequality only in this form.. $\frac{1}{6} \leqslant \theta \leqslant 6$

Type MS · Area of a triangle using half ab sin C

Answer 5 of 12 · Question 3 part a · 3 marks

2024 JUN · Paper 01 (regional R) · extract 5 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks
5(a)	**Mark both parts together**	

Type MS · Area of a triangle using half ab sin C

Answer 6 of 12 · Question 5 part b · 3 marks

2025 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

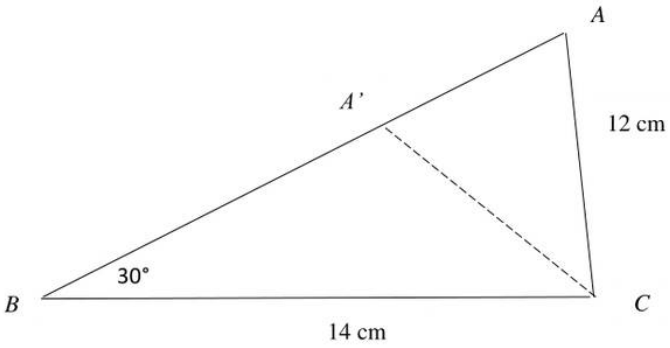
5(a)	$12^2 = 14^2 + x^2 - 2\times 14x\cos 30$	M1
	$x^2 - 14\sqrt{3}x + 52 = 0$	M1 A1
	$x = \dfrac{14\sqrt{3} \pm \sqrt{(14\sqrt{3})^2 - 4(1)(52)}}{2(1)}$	M1
	$x = 7\sqrt{3} \pm \sqrt{95}$	A1cso [5]
(b)	Area $= \dfrac{1}{2}\times 14\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \sin 30$	M1
	or uses Area $= \dfrac{1}{2}\times 12\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \dfrac{7}{12}$	
	Difference in areas	
	$D = \dfrac{1}{2}\times 14\times\left(7\sqrt{3} + \sqrt{95}\right)\times \sin 30 - \dfrac{1}{2}\times 14\times\left(7\sqrt{3} - \sqrt{95}\right)\times \sin 30$	dM1
	$D = 7\sqrt{95}$	A1 [3]
ALT Part (a) – sine rule		
	$\dfrac{12}{\sin 30} = \dfrac{14}{\sin A} \Rightarrow \sin A = \dfrac{7}{12}$	{M1
	$\left(\dfrac{7}{12}\right)^2 + \cos^2 A \equiv 1 \Rightarrow \cos A = \sqrt{1 - \left(\dfrac{7}{12}\right)^2}$	M1
	$\sin A = \dfrac{7}{12}, \cos A = \pm\dfrac{\sqrt{95}}{12}$	A1
	$x = 12\cos A + 14\cos 30 \Rightarrow x = 12\times\dfrac{\pm\sqrt{95}}{12} + 14\times\dfrac{\sqrt{3}}{2}$	M1
	$x = 7\sqrt{3} \pm \sqrt{95}$ *	A1 Cso [5]}
		Total 8 marks

Type MS · Area of a triangle using half ab sin C

Answer 6 of 12 · Question 5 part b · 3 marks

2025 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

USEFUL SKETCH



Part	Mark	Additional Guidance
(a)	M1	Use the cosine rule to form an equation. Rule to be correct but may be in any rearranged form.
	M1	Rearrange to obtain a 3TQ. Terms in any order. = 0 may be missing.
	A1	Correct 3TQ Allow poor notation – missing = 0
	M1	Attempt to solve their 3TQ by use of the formula or completing the square. See general guidance for what constitutes an attempt to solve. Use of calculators: If the 3TQ is correct, accept correct solutions without working.
	A1	Correct values for x $x = 7\sqrt{3} \pm \sqrt{95}$
(b)	M1	For use of area of triangle $= \frac{1}{2}ac\sin B$ with $a = 14$, $B = 30°$ and *their* x May use either value found for x. Or uses $= \frac{1}{2}bc\sin A$ with $c = 12$ and $\sin A = \frac{7}{12}$
	dM1	For finding the difference between their two areas provided both areas are positive This M mark is dependent on the previous M mark
	A1	Correct difference in areas (must be exact value)

	ALT Part (a) – Uses sine rule	
	M1	Use of the sine rule to find sin$A$
	M1	Use of $\sin^2 A + \cos^2 A \equiv 1$ to find an exact value for $\cos A$
	A1	Correct exact values for sin$A$ and $\cos A$
	M1	Method to find x $x = 12\cos A + 14\cos 30$ Allow *their* $\cos A$
	A1	Correct values for x $x = 7\sqrt{3} \pm \sqrt{95}$

Type MS · Area of a triangle using half ab sin C

Answer 6 of 12 · Question 5 part b · 3 marks

2025 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle using half ab sin C

Answer 7 of 12 · Question 6 part c · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

6 (a)	**Method 1** $\left(\frac{a+\sqrt{5}}{\sqrt{5}-2}\times\frac{\sqrt{5}+2}{\sqrt{5}+2}\right)=\frac{\sqrt{5}a+2a+5+2\sqrt{5}}{(1)}$ $=\left[2a+5+(2+a)\sqrt{5}\right]$ $a=3$	**Method 2** $((11+5\sqrt{5})(\sqrt{5}-2))$ $11\sqrt{5}-22+25-10\sqrt{5}$ $a=3$	M1 A1 [2]
(b)	(angle PQR =) 45° $\frac{x+3}{\sin 45^\circ}=\frac{x}{\sin 30^\circ}\Rightarrow\frac{x+3}{\frac{\sqrt{2}}{2}}=\frac{x}{\frac{1}{2}}\Rightarrow x+3=\sqrt{2}x$ $\Rightarrow 3=x\left(\sqrt{2}-1\right)\Rightarrow x=\frac{3}{\sqrt{2}-1}\times\frac{\sqrt{2}+1}{\sqrt{2}+1}\Rightarrow x=\ldots$ $x=3+3\sqrt{2}$ **OR** $\frac{x+3}{\sin 45^\circ}=\frac{x}{\sin 30^\circ}\Rightarrow\frac{x+3}{\frac{\sqrt{2}}{2}}=\frac{x}{\frac{1}{2}}\Rightarrow\sqrt{2}x+3\sqrt{2}=2x$ $x\left(2-\sqrt{2}\right)=3\sqrt{2}\Rightarrow x=\frac{3\sqrt{2}}{2-\sqrt{2}}\times\frac{2+\sqrt{2}}{2+\sqrt{2}}\Rightarrow x=$ $x=3+3\sqrt{2}$		M1 M1 A1* cso [3] [M1 M1 A1]
(c)	$\frac{1}{2}\times(3+3\sqrt{2})\times(6+3\sqrt{2})\times\sin(180-30-45)^\circ$ $\frac{1}{2}\times(3+3\sqrt{2})\times(6+3\sqrt{2})\times\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right)$ $\frac{9}{8}\left(4\sqrt{6}+4\sqrt{2}+3\sqrt{12}+6\right)$ or $\frac{1}{8}\left(36\sqrt{6}+36\sqrt{2}+54\sqrt{12}+54\right)$ oe with two out of four terms correct. $\frac{9}{8}\left(4\sqrt{6}+4\sqrt{2}+6\sqrt{3}+6\right)$		M1 dM1 A1 cao [3]
			Total 8 marks

Type MS · Area of a triangle using half ab sin C

Answer 7 of 12 · Question 6 part c · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
Throughout this question allow missing degree signs.		
(a)	M1	Any three out of four terms correct in the numerator of method 1 or in the expansion of method 2. If they use method 1 we must see the expansion come from multiplying numerator and denominator by $\sqrt{5}+2$
	A1	Cao If the answer $a = 3$ is stated without sufficient working seen award M0A0.
(b)	M1	For using a correct sine rule, either way around, using the exact values of sin 45 and sin 30, and attempting to simplify to $kx+3=\sqrt{2}x$ where k is a constant. Allow $\frac{kx}{2}+\frac{3}{2}=\frac{\sqrt{2}x}{2}$ for this mark
	M1	For factorising their $x+3=\sqrt{2}x$ [which must have come from using sine rule] to obtain $3=x\left(\sqrt{2}-k\right)$, rearranging and multiplying numerator and denominator by $\sqrt{2}+k$ This is a given answer – this step must be seen explicitly
	A1*	cso, no errors or emissions.
(c)	M1	Follow general guidance, if area formula is quoted, allow one slip in substitution. If formula not quoted, must be a fully correct substitution. For this mark only allow the angle to be $(180-30-45)^\circ$ If they work in terms of x and obtain $\frac{1}{2}\times x\times(x+3)\times\sin 105^\circ$ then do not award the first M mark until they have substituted the given value of $3+3\sqrt{2}$ for x.
	dM1	Replaces $\sin 105^\circ$ with the given value and attempts to multiply out. The mark can be awarded for any two terms correct, four terms must be present.
	A1cao	Must be in the given form with the terms in any order.

Type MS · Area of a triangle using half ab sin C

Answer 7 of 12 · Question 6 part c · 3 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle using half ab sin C

Answer 8 of 12 · Question 3 part c · 4 marks

2023 JUN · Paper 01 (regional R) · find part c in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Area of a triangle using half ab sin C

Answer 9 of 12 · Question 8 part b · 4 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

8 (a) (i)	$\overrightarrow{(AB)} = \overrightarrow{AO} + \overrightarrow{OB}$ $(= -4\mathbf{a} - 5\mathbf{b} + 8\mathbf{a} - \mathbf{b}) = 4\mathbf{a} - 6\mathbf{b}$	M1A1 [2]
(ii)	$\left\{ \left\| \overrightarrow{AB} \right\| = \sqrt{52} \right\} = 2\sqrt{13}$	B1 [1]
(b)	$\left\| \overrightarrow{OA} \right\| = \sqrt{4^2 + 5^2} \left(= \sqrt{41} \right)$ *or* $\left\| \overrightarrow{OB} \right\| = \sqrt{8^2 + 1^2} \left(= \sqrt{65} \right)$	M1
	$\left(\sqrt{52}\right)^2 = \left(\sqrt{41}\right)^2 + \left(\sqrt{65}\right)^2 - 2 \times \sqrt{41} \times \sqrt{65} \cos(AOB)$	M1
	$\left\{ \text{angle } AOB = \cos^{-1}\left(\frac{41 + 65 - 52}{2 \times \sqrt{41} \times \sqrt{65}} \right) = 58.465... \right\}$	
	Area triangle $AOB = \frac{1}{2} \times "\sqrt{41}" \times "\sqrt{65}" \times \sin "58.465..."$	M1
	$= 22$ awrt	A1 [4]
ALT 1	Area $= \frac{1}{2} \begin{vmatrix} 0 & 4 & 8 & 0 \\ 0 & 5 & -1 & 0 \end{vmatrix}$ oe	M1
	$= \frac{1}{2} \left\| (0 \times 5 + 4 \times (-1) + 8 \times 0) - (0 \times 4 + 5 \times 8 + (-1) \times 0) \right\|$ oe	M1
	$\left\{ = \frac{1}{2} \left\| -44 \right\| \right\} = 22$	M1A1 [4]
ALT 2	Use Heron's formula	
	$\left\| \overrightarrow{OA} \right\| = \sqrt{4^2 + 5^2} \left(= \sqrt{41} \right)$ *or* $\left\| \overrightarrow{OB} \right\| = \sqrt{8^2 + 1^2} \left(= \sqrt{65} \right)$	M1
	$p = \frac{"\sqrt{41}" + "\sqrt{65}" + "\sqrt{52}"}{2}$	M1
	Area $= \sqrt{p(p - "\sqrt{41}") \times (p - "\sqrt{65}") \times (p - "\sqrt{52}")}$	M1
	$= 22$ exact	A1 [4]
(c)	$\overrightarrow{OC} = \lambda \overrightarrow{OD} \left\{ = \lambda(15\mathbf{a} + 10\mathbf{b}) \right\}$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \mu \overrightarrow{AB} \left\{ = 4\mathbf{a} + 5\mathbf{b} + \mu(4\mathbf{a} - 6\mathbf{b}) = (4 + 4\mu)\mathbf{a} + (5 - 6\mu)\mathbf{b} \right\}$	M1
	$4 + 4\mu = 15\lambda$ *and* $5 - 6\mu = 10\lambda \Rightarrow \mu = ...$ *or* $\lambda = ...$	ddM1
	$\mu = \frac{7}{26}$ *or* $\lambda = \frac{22}{65}$	A1
	$\overrightarrow{OC} = \frac{66}{13}\mathbf{a} + \frac{44}{13}\mathbf{b}$ or $5\frac{1}{13}\mathbf{a} + 3\frac{5}{13}\mathbf{b}$ cao	A1 [5]
		Total 12 marks

Type MS · Area of a triangle using half ab sin C

Answer 9 of 12 · Question 8 part b · 4 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)(i)	M1	For a correct vector path to find $\overrightarrow{AB}$ or $\overrightarrow{BA}$ or for $\pm(-4\mathbf{a}-5\mathbf{b}+8\mathbf{a}-\mathbf{b})$
	A1	For the correct vector $4\mathbf{a}-6\mathbf{b}$
(ii)	B1	For the correct exact simplest magnitude $2\sqrt{13}$
(b)	M1	For the correct simplified or unsimplified $\lvert\overrightarrow{OA}\rvert$ or $\lvert\overrightarrow{OB}\rvert$
	M1	For correct substitution into the cosine rule. e.g. $(\sqrt{65})^2=(\sqrt{41})^2+(\sqrt{52})^2-2\times\sqrt{41}\times\sqrt{52}\cos(OAB)$ or $(\sqrt{41})^2=(\sqrt{52})^2+(\sqrt{65})^2-2\times\sqrt{52}\times\sqrt{65}\cos(ABO)$
	M1	For correct substitution into the area of a triangle formula, use their values. e.g. $\frac{1}{2}\times\text{"}\sqrt{41}\text{"}\times\text{"}\sqrt{52}\text{"}\times\sin\text{"}72.349...\text{"}$ or $\frac{1}{2}\times\text{"}\sqrt{65}\text{"}\times\text{"}\sqrt{52}\text{"}\times\sin\text{"}49.184...\text{"}$
	A1	awrt 22
ALT1	M1	For a correct statement for the area, must have $\frac{1}{2}$ (Start with any point, go clockwise or anticlockwise, but must repeat the first point at the end.)
	M1	For a correct evaluation of their determinant, ignore $\frac{1}{2}$
	M1	$\frac{1}{2}$ multiplies their determinant value, correct answer implies this mark.
	A1	Correct **exact** value 22, **not awrt**
ALT2	M1	For the correct simplified or unsimplified $\lvert\overrightarrow{OA}\rvert$ or $\lvert\overrightarrow{OB}\rvert$
	M1	For finding half of the perimeter
	M1	Substitute the perimeter and 3 lengths to a correct Heron's formula
	A1	Correct **exact** value 22, **not awrt**
(c)	M1	For $\overrightarrow{OC}=\lambda\overrightarrow{OD}$ (can be implied in their later work)
	M1	For correct vector statement $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{AB}$ or $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{BA}$ oe Can be implied for later substitution of their vectors **Please note**: $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{BA}$ or $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{AB}$ is also correct, provided they have found a negative value of μ
	ddM1	Equating components and reaching a value for either of their parameters. Dependent on both previous method marks
	A1	Either correct value of their parameters. They do not need to work out both values. Note: $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{AB}\Rightarrow\mu=\frac{7}{26}$, $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{BA}\Rightarrow\mu=\frac{19}{26}$
	A1	Correct vector in simplified expression

Type MS · Area of a triangle using half ab sin C

Answer 9 of 12 · Question 8 part b · 4 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle using half ab sin C

Answer 10 of 12 · Question 3 part b · 3 marks

2025 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

3 (a)	$15^2 = 12^2 + x^2 - 2\times 12\times x\times \cos 60^\circ$ $\Rightarrow 225 = 144 + x^2 - 12x \Rightarrow x^2 - 12x - 81 = 0$ *	M1 M1A1 cso [3]
	Special Case in this part for using Sine Rule **The working must be CORRECT for the award of these marks.**	
	$\frac{15}{\sin 60} = \frac{12}{\sin C} \Rightarrow C = 43.9^\circ$ **AND** $\frac{15}{\sin 60} = \frac{x}{\sin 76.1..}$ $\Rightarrow x = 16.8(16...), -4.8(16...)$ **Both roots required** $x^2 - 12x - 81 = 0 \Rightarrow x = 16.8...,\ \ 4.81...$ and concludes that x must satisfy the given quadratic equation	B1 B1
(b)	$x^2 - 12x - 81 = 0 \Rightarrow x = 6 + \sqrt{117} = \left[6 + 3\sqrt{13}\right]$ $A = \frac{1}{2}\times 12\times \left(6 + 3\sqrt{13}\right)\times \sin 60^\circ$ $A = 3\sqrt{3}\left(6 + 3\sqrt{13}\right) = 9\left(2\sqrt{3} + \sqrt{39}\right)$	M1 M1 A1 [3]
		Total 6 marks

Type MS · Area of a triangle using half ab sin C

Answer 10 of 12 · Question 3 part b · 3 marks

2025 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Area of a triangle using half ab sin C

Answer 11 of 12 · Question 3 part whole · 5 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

A1	$3y + 2x = 12$
	NB If both are correct but not in the form $ax + by = c$ then award A1A0
(b)	For all 3 marks accept < and > instead of ≤ and ≥
B1	$x \geq 0$
B1ft	$y + 2x \leq 8$ oe (ft l_1)
B1ft	$3y + 2x \geq 12$ oe (ft l_2)

Type MS · Area of a triangle using half ab sin C

Answer 11 of 12 · Question 3 part whole · 5 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks
3	$33 = \frac{1}{2} \times 11 \times 12 \times \sin \angle ABC \Rightarrow \sin \angle ABC = \frac{1}{2}$ $\left[30^\circ \text{ or } 150^\circ\right]$	M1A1
	$AC = \sqrt{11^2 + 12^2 - 2 \times 11 \times 12 \cos 30^\circ} = 6.0306... \approx 6.03 \text{ (cm)}$ $AC = \sqrt{11^2 + 12^2 - 2 \times 11 \times 12 \cos 150^\circ} = 22.2178... \approx 22.2 \text{ (cm)}$	M1A1 A1 [5]
		Total 5 marks

Type MS · Area of a triangle using half ab sin C

Answer 12 of 12 · Question 7 part b · 7 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	**Throughout this question condone missing degree signs** $\cos\theta^\circ = \frac{6^2+8^2-k^2}{2\times6\times8} = \frac{100-k^2}{96}$ *	M1A1cso [2]
(b)	$\sqrt{455} = \frac{1}{2}\times6\times8\times\sin\theta^\circ$ $\Rightarrow \sin\theta^\circ = \frac{\sqrt{455}}{24} \Rightarrow \left(\sin^2\theta^\circ = \frac{455}{576}\right)$	M1 A1
	$\cos^2\theta^\circ = 1-\sin^2\theta^\circ \Rightarrow \cos^2\theta^\circ = 1-\frac{455}{576} = \frac{121}{576}$ $\Rightarrow \cos\theta^\circ = \pm\frac{11}{24}$ both values required $\cos\theta^\circ = \frac{11}{24} = \frac{100-k^2}{96} \Rightarrow k^2 = 56 \Rightarrow k = \sqrt{56} = \left(2\sqrt{14}\right)$ $\cos\theta^\circ = -\frac{11}{24} = \frac{100-k^2}{96} \Rightarrow k^2 = 144 \Rightarrow k = 12$	M1 A1 M1A1 A1
	ALT $\sqrt{455} = \frac{1}{2}\times6\times8\times\sin\theta^\circ$ $\Rightarrow \sin\theta^\circ = \frac{\sqrt{455}}{24}$ $\theta^\circ = \sin^{-1}\left(\frac{\sqrt{455}}{24}\right)$ $(= 62.72\ldots^\circ)$ $\theta^\circ = 62.72\ldots, 117.27\ldots$ both values required $\cos\theta^\circ = \frac{100-k^2}{96} \Rightarrow k^2 = 100-96\cos\theta^\circ$ $\Rightarrow k^2 = 100-96\cos 62.72\ldots \Rightarrow k^2 = 56 \Rightarrow k = \sqrt{56}$ $\cos\theta^\circ = \frac{100-k^2}{96} \Rightarrow k^2 = 100-96\cos\theta^\circ$ $\Rightarrow k^2 = 100-96\cos 117.27\ldots \Rightarrow k^2 = 144 \Rightarrow k = 12$	[7] [M1 A1 M1 A1 M1A1 A1]
		Total 9 marks

Type MS · Area of a triangle using half ab sin C

Answer 12 of 12 · Question 7 part b · 7 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

(b)	**M1**	For using the **correct** formula for area of a triangle and substitution of the given values to obtain $\sqrt{455} = \frac{1}{2} \times 6 \times 8 \times \sin\theta°$ and attempt to rearrange to obtain $\sin\theta° = \cdots$
	A1	$\sin\theta° = \frac{\sqrt{455}}{24}$ Allow for $\sin\theta = \frac{\sqrt{455}}{0.5\times8\times6}$
	M1	For use of $\sin^2\theta + \cos^2\theta = 1$ to obtain a value for $\cos\theta°$. $\cos^2\theta° = 1 - \frac{'455'}{576} \Rightarrow \cos\theta° = \pm\sqrt{1 - \frac{'455'}{576}}$ Allow use of their $\sin\theta°$ provided $-1 \leq \sin\theta° \leq 1$ Allow if only one value of $\cos\theta°$ obtained.
	A1	$\cos\theta° = \pm\frac{11}{24}$
	M1	For forming an equation for k using their $\cos\theta°$ and attempt to solve for k. $\cos\theta° = \frac{'11'}{24} = \frac{100-k^2}{96} \Rightarrow k^2 = 56 \Rightarrow k = \sqrt{56}$ $\cos\theta° = '-\frac{11'}{24} = \frac{100-k^2}{96} \Rightarrow k^2 = 144 \Rightarrow k = 12$
	A1	For one correct value of k $\sqrt{56}$ or awrt 7.48 **or** awrt 12
	A1	For both correct values of k $\sqrt{56}$ or awrt 7.48 **and** awrt 12
ALT – working with angles		
	M1	For using the **correct** formula for area of a triangle and substitution of the given values to obtain $\sqrt{455} = \frac{1}{2} \times 6 \times 8 \times \sin\theta°$ and attempt to rearrange to obtain $\sin\theta° = \cdots$
	A1	$\sin\theta° = \frac{\sqrt{455}}{24}$ Allow for $\sin\theta = \frac{\sqrt{455}}{0.5\times8\times6}$
	M1	For use of the inverse trigonometric function to obtain a value for $\theta°$ $\theta° = 62.72\ldots, 117.27\ldots$ If working not shown then award for angle correct to a minimum of 1 d.p. Allow if only one value of $\theta°$ found. Condone working in radians awrt 1.09, awrt 2.05
	A1	$\theta° = 62.72\ldots, 117.27\ldots$ Allow awrt 62.7°, 117.3° Both angles found. Condone working in radians awrt 1.09, awrt 2.05
	M1	For forming an equation in k using their θ and an attempt to solve for k. $\cos\theta° = \frac{100-k^2}{96} \Rightarrow k^2 = 100 - 96\cos\theta°$ $\Rightarrow k^2 = 100 - 96\cos 62.72\ldots \Rightarrow k^2 = 56 \Rightarrow k = \sqrt{56}$ $\cos\theta° = \frac{100-k^2}{96} \Rightarrow k^2 = 100 - 96\cos\theta°$ $\Rightarrow k^2 = 100 - 96\cos 117.27\ldots \Rightarrow k^2 = 144 \Rightarrow k = 12$
	A1	For one correct value of k $\sqrt{56}$ or awrt 7.48 **or** awrt 12
	A1	For both correct values of k and no others. $\sqrt{56}$ or awrt 7.48 **and** awrt 12

Type MS · Area of a triangle using half ab sin C

Answer 12 of 12 · Question 7 part b · 7 marks

2021 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks